Investigating How Designers Perceive the Timing and Content of Proactive AI-generated Feedback

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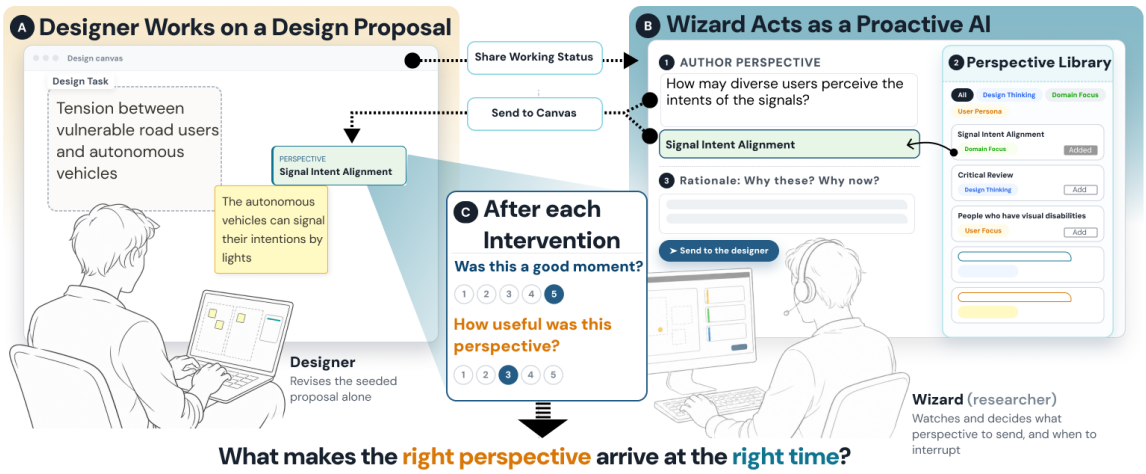


Fig. 1. Our Wizard-of-Oz system for studying proactive AI perspective-taking in interaction design. A wizard observes a designer's evolving work, composes and sends perspectives in situ, and documents why each intervention was chosen and sent at that moment. Designers receive the intervention and then rate its timing and usefulness, enabling us to study the interplay between what perspective is offered and when it is sent.

Designers increasingly ask generative AI for feedback on creative artifacts. What is the potential for AI to proactively guide and spark ongoing work? Such proactivity creates an orchestration problem: what to offer, how to frame it, and when to inject it into the process. To explore this, we created a Wizard-of-Oz system where designers receive in-situ AI-generated perspectives, mediated by a human wizard. In a study with N=20 designers, the wizard generated a total of 81 proactive interventions, including design thinking strategies and synthetic stakeholder perspectives. Participants rated each intervention in real time for its perceived timeliness and usefulness. These judgments frequently diverged: useful content could arrive at bad moments, while good timing did not guarantee a useful perspective. We identify when and why these perspectives provide value, as well as why even carefully selected perspectives can miss designers' needs. Our findings offer implications for future autonomous proactive design support.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; **Interactive systems and tools**; *HCI design and evaluation methods*; • **Computing methodologies** → *Artificial intelligence.*

Additional Key Words and Phrases: Wizard-of-Oz, mixed-initiative system, proactive AI, interaction design

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1 INTRODUCTION

Designers use external feedback to evaluate and refine their work [42, 58]. The critique session exists because designers can become anchored to their early framings and need someone outside the work to unsettle them [33, 42, 52]. Creativity-support research has accordingly treated computers both as tools that enable communication and as collaborators that give feedback [20, 41, 55]. Large language models (LLMs) can now play that outside role on demand. A model can be prompted into the position of a stakeholder, or into the skeptic who asks what breaks first [11, 40, 46, 54]. On-demand feedback, however, still requires designers to initiate a request, including when they may not recognize which perspective is missing. Proactive support could surface such perspectives as the work evolves, but must decide which contributions warrant the designer's attention and when to offer them.

Imagine a good colleague, someone who not only has the right thing to say, but also knows the right moment to say it. Designing proactive artificial intelligence (AI) raises the same two questions: what content should the system contribute, and when should it step in? Recent work has increasingly explored the first content question by building AI agents that enact perspectives a designer may not hold [4, 21, 23]. These include LLMs cast as devil's advocates [11, 26], persona agents that provide feedback from synthetic audiences or stakeholders [36, 54], and multi-agent systems that let users choose which expert perspectives are enacted during ideation [40, 46, 48]. A parallel line of work examined the timing question of when proactive agents should intervene [34, 59]. For example, Johnson et al. found that teams wanted proactive agents to avoid interrupting ongoing conversation and instead enter during natural pauses [34]. In open-ended design work, content and timing are intertwined as designers' ideas and needs evolve. A perspective may be premature at one stage and useful at another, while relevant feedback can arrive when the designer is not ready to engage. Proactive support therefore requires a joint decision about when to intervene and which perspective to surface.

We call this problem ***proactive feedback orchestration***. It treats each AI-generated perspective as a candidate intervention and asks whether, when, and in what form it should reach the designer. This is a mixed-initiative problem because the cost of an ill-judged contribution falls on the person whose work it interrupts [30]. Prior studies have identified how suggestion content and timing shape users' experiences of proactive assistance [25, 47]. What remains less understood is how the judgments behind selecting and timing a perspective align with designers' readiness to receive it and the usefulness they find in it as their work evolves. Acceptance or dismissal alone cannot distinguish these dimensions, while records of feedback alone omit decisions to delay or withhold a candidate.

Building on work that tries to make decision rationales in creative workflows explicit [56], we investigate proactive feedback orchestration by putting a human wizard in the orchestrator seat and studying the decisions they made. In our Wizard-of-Oz system, a designer revises an existing proposal on a canvas with a reactive AI assistant they can use. From another room, a wizard watches a live mirror of the canvas, considers how to introduce AI-enacted perspectives, composes them, and sends or queues them based on their judgment of the designer's ongoing work. After each intervention, the designer first rates *the moment* before seeing the feedback. Then they rate *the content* after reading it, allowing timing and usefulness to be recorded separately and in the moment. The system also records the interaction trace, the wizard's rationale, and how they send the perspectives.

This paper asks three questions:

- **RQ1. What makes a proactive AI-enacted perspective useful in ongoing design work?** To study this, we created a Wizard-of-Oz system in which a human wizard was augmented to generate AI-enacted perspectives for giving feedback. In a study with $N = 20$ designers, we identified forms of value that proactive AI can provide even

when designers already have access to a reactive AI assistant, such as foregrounding overlooked stakeholders and scaffolding designers' thinking processes (Section 5.1).

- **RQ2. When is a proactive perspective welcome, and what signals of that receptivity can we learn?** To answer this question, we paired the wizard's in-situ reasoning about whether to intervene with designers' evaluation of the timing of those interventions (Section 5.2).
- **RQ3. What design considerations emerge for future autonomous proactive design support?** Part of the answer came from occupying the wizard's seat ourselves—examining what the wizard needed at hand, which strategies we believed would be useful, and how those strategies came across in practice (Section 3.1). The rest comes from how designers envisioned proactive interaction after experiencing such mixed-initiative collaboration (Section 5.3).

Across 20 design sessions and 81 proactive interventions with our Wizard-of-Oz system, the study contributes:

(1) **an empirical understanding of proactive feedback orchestration**, characterizing what designers find useful in AI-enacted perspectives, showing how timing and usefulness judgments can diverge, and organizing explanations of receptivity across session stage, task frame, and attention availability.
(2) **design implications for autonomous proactive design support**, including interpreting activity cues in context, making the intervention negotiable when receptivity is uncertain, and distinguishing perspectives that broaden designers' thinking from those that develop or confirm existing ideas.

2 RELATED WORK

2.1 Multi-perspective design support

Design work benefits from external perspectives because designers become anchored to familiar examples and their own assumptions [33, 52]. Research on design inspiration likewise shows that examples are integral to practice but can narrow exploration, and that designers draw from a limited subset of the sources available to them [24, 27, 57]. Structured techniques make the shift of attention deliberate. The Six Thinking Hats encourages participants to take on six different perspectives, each represented by a hat: the White Hat focuses on facts, the Red Hat on feelings, the Black Hat on risks, the Yellow Hat on benefits, the Green Hat on creativity, and the Blue Hat on the thinking process [14]. Multiperspective representations counter fixation by exposing designers to alternative views of the same object [53]. Researchers have also explored structured ways to collect and organize diverse design feedback. CrowdCrit structures the task of design critique for crowd workers and aggregates their feedback, while UICrit collects design critiques from experienced designers and characterizes the design issues they cover [16, 42]. A perspective can thus redirect attention toward a concern the designer might otherwise miss.

A growing body of research explores AI-supported perspectives based on thinking strategies, personas, and domain expertise. Devil's advocate agents challenge assumptions in group decision-making [11], while a Wizard-of-Oz study explored metacognitive support for reflection on goals and evaluation strategies during co-creation [23]. For design critique, PosterMate simulates audience personas for poster design and reconciles their feedback through a moderator agent [54]. Criticmate structures UI critique into editable situation-awareness stages [38], while CritiqueCrew coordinates expert roles such as UX, product, and engineering to critique designs in Figma [10].

Related work explores multi-perspective interaction in decision-making, professional ideation, and research. ChoiceMates uses role-diverse agents for unfamiliar decision-making and improved situation understanding and decision confidence compared with web search [46]. MultiColleagues uses role-diverse agents for structured professional ideation

and produced ideas rated higher in quality and novelty than a single-agent baseline [48]. Perspectra lets researchers choose expert agents, branch discussions, and inspect argument structure during proposal ideation [40]. In-situ thought exchanges support critical paper reading in single-agent and multi-agent configurations [19].

Making multiple perspectives available also raises questions about how they should enter ongoing work. In speculative workshops on collaborative GenAI, Johnson et al. [34] found that participants wanted control over agent involvement, including options for agents to signal that they had a contribution and wait to be invited. Our probe treats each generated perspective as a candidate intervention that may be surfaced, queued, revised, or withheld as design work unfolds. We examine which perspectives warrant the designer's attention and when they should be offered.

2.2 Timing and content in proactive assistance

Mixed-initiative systems require decisions about when the system should act and when it should defer to the user. Foundational work on mixed-initiative interaction argues that effective systems reason about the potential costs of taking initiative, such as bothering the user, and the potential benefits, such as reducing user effort to accomplish a task [30]. Interruption research distinguishes immediate, negotiated, mediated, and scheduled coordination [44], and shows that interruptions imposed within a task can cost more in performance and affect than the same interruption delivered at a task boundary [1, 5]. Interactive machine learning emphasizes giving users ways to steer, correct, and interpret model behavior to guide models toward desired outcomes [2]. Human-agent collaboration frameworks highlight that agents should function as useful collaborators rather than distracting autonomous actors, with commitment, responsiveness, and support as key qualities for effective collaboration [12].

Timing assistance requires judging the receiver's cognitive openness: whether the designer is ready to consider an outside perspective at that moment. Just-in-time adaptive interventions formalize the problem of deciding when a person is receptive to support [37], and interruption studies show that users' recovery from interruptions is influenced by their attention to the task before suspension [32], highlighting the importance of timing when interrupting ongoing work. Context-aware systems have found activity transitions to be less burdensome delivery points than random moments [29]; however, in-situ notification studies also show that receptivity depends jointly on task state, presentation, social relation, and individual differences [45]. Proactive programming support illustrates why visible activity may be an unreliable timing cue. Pu et al. [47] note that idleness can reflect ongoing thought and report that some participants disliked interventions triggered by inactivity. WoZ probes of cues used to guide collaborative ideation show that wizards attend to pauses, conversational specificity, and pace when sending cues to ideators, yet some cues can be missed or ignored when the timing is inappropriate [49].

Content can also fail to meet users' needs. Gu et al. [25] found that some analysts regarded suggestions as too basic, while others struggled to understand suggestions beyond their statistical background. In *Need Help?*, participants criticized proactive suggestions whose relevance to their current programming task was unclear [9]. Timing and content can also interact. Gu et al. [25] found that analysts considered some suggestions unhelpful at arrival but potentially useful at another point in their work. This motivates examining what assistance contributes in context alongside whether its arrival is welcome.

2.3 Wizard-of-Oz as a data-collection method

Wizard-of-Oz (WoZ) lets researchers test an interaction before its underlying behavior is automated and collect examples that can inform that automation [13, 22]. Early applications used simulated system behavior to iteratively develop natural-language interfaces and intelligent agents before their automation was feasible [35, 43]. For example, Gu et al.

[25] used an LLM-assisted wizard to control the content and timing of suggestions in analysts' notebooks. Gmeiner et al. [22] combined model-generated support with human control over intervention timing and correction of system outputs during live design work. These arrangements let researchers test generated assistance in ongoing work while retaining control over how it enters the interaction.

Recording why decisions were made provides evidence beyond the outputs alone. ClearFairy links creative actions and artifacts with elicited explanations and inferred rationales; exploratory applications showed that these records could improve the alignment of generated next-step suggestions with professionals' intentions [56]. This motivates recording why an intervention was considered, rather than only what was shown. Counterfactual replay offers a way to examine alternative model responses against recorded work contexts [22]. Such replay cannot establish how an alternative intervention would have changed the designer's subsequent work or receptivity.

Here, we use WoZ to examine designers' responses to proactive perspectives alongside the decisions that produced them. Our study records candidate perspectives, work context, and the wizard's reasons for sending, queuing, revising, or withholding them. For the perspectives that were sent, we link these records to designers' timing and usefulness judgments when available. This lets us examine how the wizard's reasons for intervening relate to the designer's experience. We use these records to inform future automation, treating wizard decisions as situated judgments rather than ground-truth labels for an automated policy.

3 A WIZARD-OF-OZ SYSTEM FOR STUDYING PERSPECTIVE ORCHESTRATION

We approached perspective orchestration through iterative prototyping, using the evolving artifact as a means to frame and investigate an under-constrained interaction problem [60]. Our first design probe gave designers a fixed set of perspectives to draw on during a design task, and let them author new ones. The probe surfaced a tension that a fixed set might not fit evolving ideas, yet authoring a new perspective was itself taxing.

To approach this challenge, we moved to a mixed-initiative Wizard-of-Oz setup in which a human wizard composes perspectives and decides what and when to introduce them. The designer, meanwhile, works only with a general-purpose reactive AI that they can consult on demand. Holding the two apart lets us study what a proactive perspective adds when the designer already has an AI assistant at hand, and how the wizard composes and times each one. Below, we present five prototypes from this iterative process, summarized in Figure 2.

3.1 Iterative system design process

V1 *The designer works directly with multiple perspectives.* To understand what makes perspective orchestration difficult, V1 began with the most direct interaction model for multi-perspective support. The system has a sidebar that presents a set of perspectives, and the designer decides which one to view. This provided a simple baseline for asking whether access to diverse viewpoints was sufficient before introducing proactive selection or timing. Inspired by the six thinking hats approach [14], we deployed six thinking-strategy cards such as inspiring thinking and critical thinking, and let designers author new perspective cards of their own.

Pilot testing on this probe surfaced three problems that perspective orchestration has to address:

(1) A fixed set of perspectives may not fit ideas that are evolving.
(2) Yet authoring a new perspective is taxing, and it pulls attention away from the design task.
(3) The interface for perspectives, such as a dedicated sidebar, competes with the generic AI-assistant sidebar designers already expect.

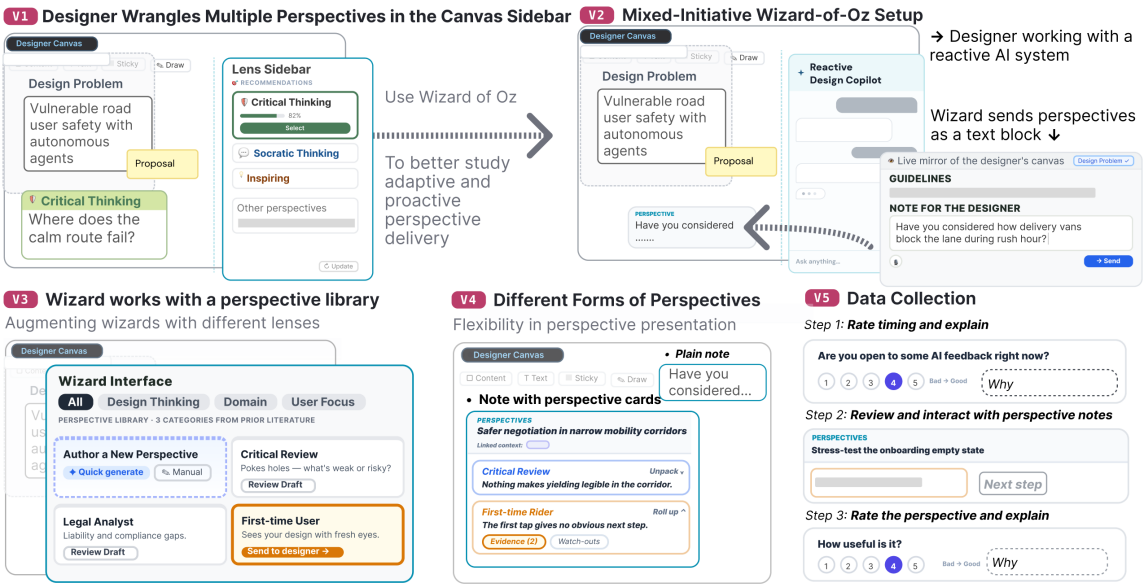


Fig. 2. Overview of our five-stage iterative system design process. We began with designer-managed perspective selection (V1), introduced a Wizard-of-Oz setup (V2), augmented the wizard with a structured perspective library (V3), supported flexible and context-linked interventions (V4), and finally embedded in-situ measures of intervention timing and usefulness (V5).

This motivates us to study perspective orchestration toward proactivity that takes the selection and its timing off the designer's hands.

We therefore adopted a Wizard-of-Oz setup to observe those judgments being made before committing to a policy that automates them.

V2 *A wizard writes perspectives while the designer works with reactive support.* V2 introduced a human wizard to occupy the proactive orchestration seat. To see how a wizard would approach perspective support without prescribing how, we let the wizard write each perspective as a plain-text message: they monitored the designer's canvas and sent one perspective at a time. On the designer's side, we kept a reactive AI assistant that responded only to explicit requests. The wizard's role was therefore to anticipate perspectives the designer might find useful and to judge whether they warranted an interruption.

However, pilot use then showed that the *wizard also needed support for perspective orchestration.* Watching the designer's unfolding work while composing a grounded perspective was hard, and when the wizard wanted to enact another stakeholder's angle, writing the whole message from scratch left it without evidence or illustration. This motivated V3: augmenting the wizard.

V3 *A perspective library augments the wizard.* V3 augmented the wizard interface with a library of perspective cards in three categories, each grounded in the prior work on AI-enacted perspectives reviewed in Section 2.1: *design thinking*, for shifts in thinking mode [14, 53]; *domain expertise* (labeled Domain in the interface), for discipline-specific judgment [10, 40]; and *user persona* (User Focus in the interface), for a stakeholder absent from the room [26, 54]. The wizard can

also generate new cards and new categories of their own, so the supply is not limited to the pre-introduced categories. In this stage, the wizard can send these perspective cards or write their own notes like in V2 .

From the pilot test, we learned that the richer supply eased composition but exposed a presentation problem. Sending the cards directly, such as a certain user persona, left the designer to work out how to interpret and link them. We learned that not only which perspectives are sent but also the form in which they are presented matters.

V4 *Support flexible and contextual orchestration.* V4 expanded and unified how perspectives are presented and operated on. The wizard could compose a *perspective node* as either a plain message or a note containing one or more generated perspective cards. In both forms, the feedback shown to designers was generated or revised with the language model, then selected, assembled, or edited by the wizard before intervention. We use *plain note* to describe a presentation form without attached perspective cards, rather than human-only authorship. The library stays available as a place to look for an angle or to help draft the note. The same interface can present from a one-line nudge to a themed bundle of perspectives, so the wizard can fit the form of an intervention to what the moment calls for.

For perspective node content generation, we also added a context link function where the wizard can point the generation at a specific text block on the designer's canvas, or at a frame that contains multiple blocks. Once landed on the designer's canvas, the node also links back to what it was drawn from, to indicate the connection. The wizard can also send without linking anything, in which case the generation draws on the whole canvas.

V5 *In-situ judgments of timing and usefulness.* By V4, the environment had accumulated the touch points for perspective orchestration, such as what the wizard composed, the context it was written in, and when to introduce the perspectives. V5 turned those touch points into traces. We asked designers two questions about the intervention — one about the moment, shown before the content is revealed, and one about how useful the content is. In this way, timing and usefulness are rated separately, each in situ.

Five structured pilot sessions stabilized the instrument and the protocol. Figure 3 shows the final system.

3.2 The final interaction and data flow

The final system couples a designer workspace and a wizard console around the same live session. Figure 3 shows both views in use. Below, we follow one intervention from shared context to the designer's evaluation.

Establish shared task context. The designer works on a *Design Proposal* in response to a *Design Problem* on the canvas (D2). They add content blocks, sticky notes, and frames from the toolbar (D1), draw, and search the web from the bar above the canvas. A reactive AI assistant in the sidebar (D3), the *Design Copilot*,[1] offers critique, ideas, and image generation, but only when prompted. On a separate machine, the wizard sees a live mirror of the canvas and the designer's recent activity.

Compose and send a perspective node. The wizard browses and assembles *perspective cards* in the perspective library (W1). The library holds pre-authored cards (W2) in the three categories from V3 , and the wizard can author new categories. *Quick generation* drafts additional cards from the designer's board, where the wizard types a direction, optionally selects nodes in the mirror, and the model writes cards grounded in the design context. After previewing cards for inspiration, the wizard moves any number of them into the send interface (W3). There, the wizard writes a

[1]We refer to the reactive AI assistant as the Copilot, its name in the system, throughout the paper.

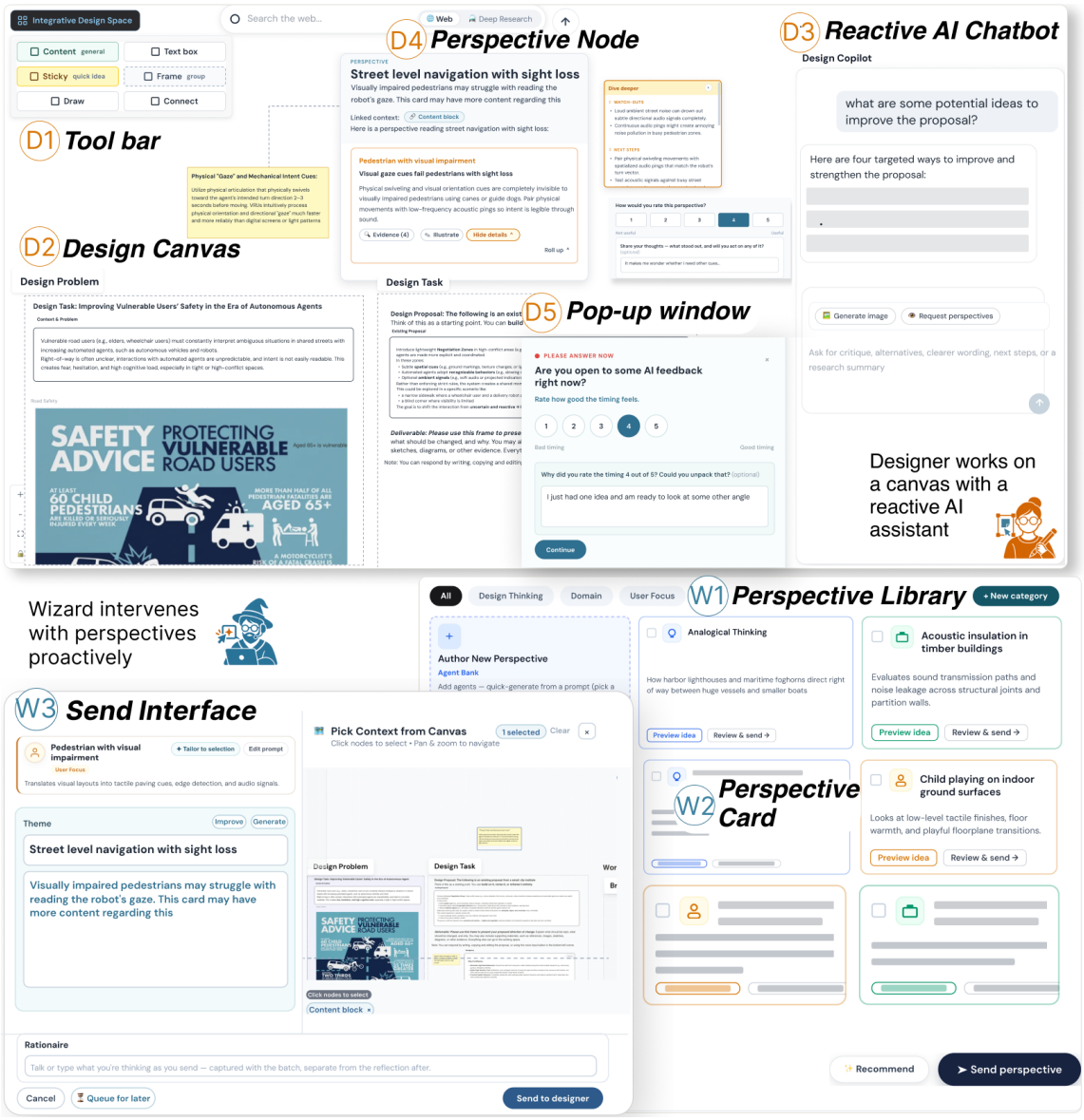


Fig. 3. The final system. Above, the designer works on a canvas (D2) with a toolbar (D1) and a reactive AI assistant, the Copilot, that responds only when asked (D3). Below, the wizard browses perspective cards (W2) in the perspective library (W1), then grounds an intervention in canvas context, gives it a theme, and sends or queues it, with or without cards, in the send interface (W3). A sent intervention first raises a pop-up that asks about the moment before showing any content (D5); once placed, it sits on the canvas as a perspective node (D4).

theme and a short summary for the node, or has the model generate or improve them from the chosen cards, and picks from the mirror the canvas nodes the perspective responds to, which become the node's *linked context*.

The wizard then either sends the node, which raises a pop-up on the designer's screen, or queues it for a better moment. Queued nodes can later be sent or discarded. On sending, a required prompt asks the wizard *why these perspectives, and why now*, specifically why this content suits this designer, and why this moment suits the intervention. The wizard answers by typing or recording a voice note at the bottom of the send interface.

Receive and evaluate the perspective node. The node first reaches the designer as a small pop-up at the corner of the canvas (D5). Before showing any title or content, the pop-up asks whether the designer is open to AI feedback right now and requires a timing rating from 1 (bad timing) to 5 (good timing), with a typed or spoken reason. It then reveals the theme and summary, and the designer places the node on the canvas, saves it for later, or dismisses it.

Once the node is placed (D4), its *linked context* control navigates back to the canvas node the perspective was grounded on, and each card shows a brief takeaway with controls for revealing evidence, suggested next steps, and a generated illustration. The node's footer asks the designer to rate the perspectives from 1 (not useful) to 5 (useful) and to say why, typed or spoken. The designer can also ask for perspectives with the *Request perspectives* tag in the Copilot (D3). The pop-up then skips the timing rating and opens directly at the theme and summary, and the request is logged as designer-initiated.

3.3 Implementation

The environment is a web application built with a React client with a React Flow canvas over a Python Flask service. Perspective generation, the recommendation pass, and the Copilot's replies are model calls to Gemini 3.6 Flash. Illustrations use GPT-image-2. Firestore holds each session and joins the two machines. The designer's client writes its board state and recent activity to the session document, and the wizard's client renders that, so the mirror stays current without a direct connection between them. We will release the code and the console upon acceptance.

4 USER STUDY

With the instrument, we ran a Wizard-of-Oz study in which designers refined a proposal for one hour while a human wizard orchestrated AI perspectives from a separate machine.

4.1 Participants and wizards

We recruited 22 designers from Information Science and design departments for one session each. Participant identifiers were assigned sequentially during recruitment and range from P1 to P22. The session recording was missing for P1, and P5 did not complete their session and was excluded. The twenty sessions with complete data (P2–P4 and P6–P22) are therefore the ones analyzed throughout. Their mean age was 26.4 years (SD = 4.2, range 19–35). Their reported design experience averaged 5.4 years (SD = 2.4). Appendix A reports the individual self-described backgrounds. Participants were compensated $30 for their participation, and the study was approved by our institution's IRB.

Two members of the research team, both design researchers, served as wizards and ran 15 and 5 sessions, respectively. They attended the pilot sessions together, practiced running the study, and co-developed a written manual (a fixed introduction, intervention procedure, disclosure boundary, and recording checklist) that they used to check that they applied the protocol consistently.[2] The second wizard thus served as a check that the judgment-based protocol could be carried out by another operator and that the findings did not rest on one wizard's practice. Each study session, however, was run by a single wizard, so inter-wizard agreement was not tested.

[2]The full wizard manual is in the supplementary material.

4.2 Task and materials

Participants worked on a proposal revision task for communicating the intent of autonomous agents such as self-driving vehicles and delivery robots to vulnerable road users such as people with disabilities, older adults, and children. We chose the domain because it is open-ended and socially rich, yet everyone has lived experience of sidewalks and streets, so participants could connect to it easily. It also rewards stakeholder reasoning and reframing, so perspectives from several directions are plausibly useful and no single participant is expected to hold them all.

Participants were not asked to generate a concept from scratch. Each session opened on a seeded board holding a *Design Problem* frame and an existing *Design Proposal*, and the task was framed as revision: "imagine this is something a colleague worked on, and then they left." Participants were told they could enhance the proposal, trim the parts they judged weak, or reframe it entirely, and that we were interested in their thinking process rather than the final artifact. This framing does two things the study depends on. It places every participant at the same starting point, so orchestration decisions across sessions are made against comparable board states. And it puts participants in *refinement* rather than ideation, which is where an outside perspective has a concrete idea to act on. A designer starting from scratch may spend the early part of the session producing material, while a designer revising an existing proposal already holds a position where an unsolicited perspective can play a role. The full task description is provided in Appendix B.

All sessions ran in the browser with the participant sharing their screen over a video call. The call was recorded in addition to the in-app instrumentation, and the orchestration side is carried by the wizard's transcribed narration together with the interaction trace of every send, queue, and cancel.

4.3 The wizard protocol

The wizard's job was to watch the designer work and decide whether to send a perspective, which one, and when. The wizard had access to the board and to the designer's spoken think-aloud if they chose to do so. Their rationales, hesitations, overrides, and misses are therefore data about the orchestration problem rather than ground truth.

The wizard's only channel was the perspective node. All perspectives, regardless of source, shared the same composition and sending flow (Section 3.2). The wizard controlled the timing and the content, not the appearance of the perspective card, following the guidelines for Wizard-of-Oz studies in [51].

The designers knew AI feedback would arrive, and they did not know it was selected by a wizard. The wizard was muted while composing, so the designer received no signals from the wizard at all. In this way, they could never sense the timing decision, and every arrival appeared to come from an AI.

4.4 Procedure

Each session ran about 85 minutes in three phases, with the refinement phase scheduled for roughly an hour.

1. Introduction and guided tour (~10 minutes). The wizard read a fixed introduction covering the design problem, the revision framing, the canvas and Copilot, voice input and microphone consent, and then walked the participant through the interface and a practice intervention.

2. Design refinement (the main session). Participants worked on the proposal using the canvas and the Design Copilot as they saw fit, thinking aloud (narrating a new thought or decision as it happened rather than keeping up continuous commentary), following concurrent verbal-report methods for capturing task-level reasoning [18].

Table 1. Data captured per session and the research questions each source informs. In-flow instruments are part of the working interface; the wizard trace and interaction log are recorded continuously.

Source	Captured	RQ
Designer in-situ ratings	Timing rating 1–5, with a follow-up reason	RQ2
for the interventions	Usefulness rating 1–5, with a follow-up reason	RQ1, RQ3
Designer post-session interview	Retrospective reflection for each intervention	RQ1, RQ3
	Open discussion on proactive AI design support	RQ3
Wizard	A rationale per intervention (*why these, why now*)	RQ1, RQ2
	Traces on how they compose each intervention	RQ1, RQ2
	Reflections for each session	All RQs
Process traces	Canvas and Copilot activity, with board snapshots	RQ2
	The recorded session call and its transcripts	RQ1–RQ3

Wizard-initiated perspectives arrived unannounced through the arrival prompt, node, and rating controls described in Section 3.2, so each intervention yielded a timing rating and reason before its content was shown and a usefulness rating and reason after. Participants could also request a perspective by describing the help they wanted; these designer-initiated requests skipped the timing prompt, opened directly to the content, and were logged separately by origin. The analyses reported below concern the 81 wizard-initiated interventions.

3. Closing questions and debrief (∼15 minutes). The participant then revisited each intervention in arrival order, with the canvas restored to its state when it arrived. For each, we asked how far they agreed with the statement *"The AI perspectives made me realize points I might not have reached on my own, even given more time"* to measure the novelty of the perspective, followed by an open question.

We closed each session with an open discussion about participants' overall experience with proactive perspectives, moments when additional perspectives might have been useful, and what kinds of proactive AI support they would find valuable.

4.5 Measures and analysis

Table 1 categorizes the data we collected and the correspondence to the research questions. We used descriptive statistics to summarize intervention frequency, category and form, ratings, dispositions, and queue and cancel decisions. Timing and usefulness were treated as paired judgments of the same intervention.

For the qualitative data, the two authors who served as wizards jointly developed an inductive codebook through iterative readings of designers' timing reasons and usefulness reflections, the wizards' rationales and narration, and the closing responses and session transcripts, drawing on thematic analysis as a flexible method for identifying patterned meaning [6]. They then applied the codebook. Disagreements in coding were discussed and resolved through consensus. An external researcher reviewed the codebook and a sample of its application to coded excerpts as an additional interpretive check. Finally, we examined the coded accounts alongside the corresponding ratings and interaction traces to contextualize the qualitative patterns. The codebook and coded excerpts are released with the study materials.

5 FINDINGS

Across 20 sessions, the wizard initiated 81 interventions ($M = 4.05$ per session, $SD = 0.76$). We first examined what proactive perspectives the wizard sent and what designers found useful from the perspective (RQ1) and whether the perspectives were brought to the designer at good or bad moments (RQ2). We then considered the results on perspective content and timing and the wizard logs to understand what decisions an automated orchestrator would need to make to provide good, well-timed perspectives (RQ3).

5.1 RQ1: What made a proactive perspective useful

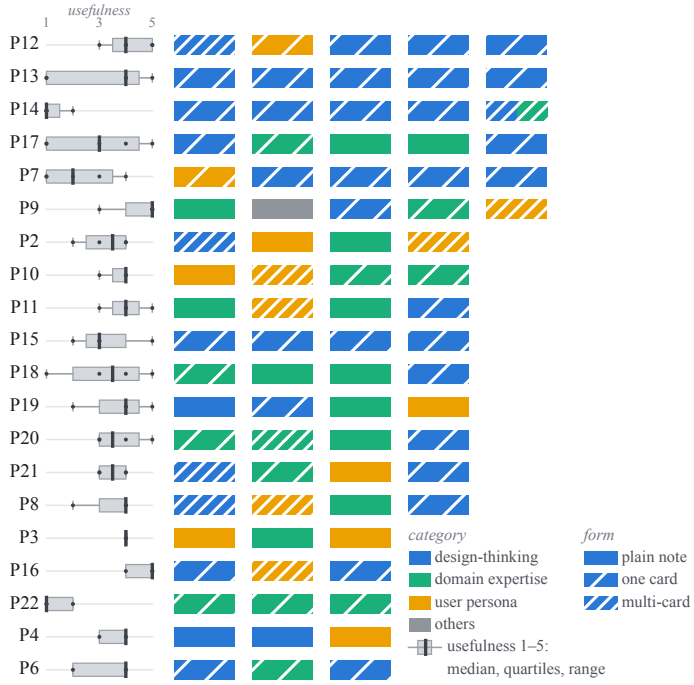


Fig. 4. The category and form of each wizard-initiated intervention, in send order. One intervention is shown in grey: a plain note that fit none of the three categories (P9). The adjacent box plots summarize usefulness ratings by session.

5.1.1 What the wizard sought to accomplish with perspective interventions. Across the sessions, the wizard initiated roughly one intervention every 15 minutes. The wizard's post-session reflections suggest that this cadence may have emerged because designers needed time to develop their ideas and may already have been using reactive AI for brainstorming, while the wizard needed time to observe the evolving work, identify a useful angle, and compose a grounded intervention.

Coding the wizard rationales recorded for sending the 81 interventions yielded four recurrent purposes. The most common was to **augment the design thinking** (16 of the 20 sessions): the wizard wanted to provide a structured thinking frame, help ideation converge, or open alternatives before premature narrowing. The wizard also **pushed against what the designer already had** (15/20), for example by identifying a failure mode, criticizing the proposal at a checkpoint, or surfacing a constraint that the idea had assumed away. A third purpose was to **bring in what**

the design proposal left out (14/20), such as unnamed critical users or other stakeholders of the system. Finally, 11 interventions across eight sessions **responded to an idea the designer was brewing**, either by deepening an emerging idea or responding to an explicit question from the designer.

5.1.2 How the wizard composed perspective interventions. Regarding how the wizard composed these interventions, Figure 4 shows that these choices formed **no fixed sequence across sessions** — no two of the 20 sessions received the same sequence of categories and presentation forms. Seventeen sessions drew on at least two categories, and 13 mixed at least two forms. Design-thinking perspectives were used most often (n=40), followed by domain-expertise (n=25) and user-persona perspectives (n=15). The wizard's rationales suggested that interventions aimed at steering the process (e.g., reframing the proposal) used a design-thinking perspective. User-persona interventions almost always named a stakeholder not included on the board (14 of 15). Domain-expertise interventions split between stress-testing the proposal (10 of 25) and bringing in another practitioner, such as an inclusive designer or a discipline the board had left out (9 of 25).

The wizard also changed the composition of a perspective presentation considering the information they wanted to convey. A single perspective card was the most often chosen (n=46, 17 sessions). Of the 12 multi-card interventions, six placed several personas in parallel. For example, P16 received a perspective node listing cards about an older adult with low vision, a deaf pedestrian, and a parent pushing a double stroller. Four multi-cards offered a set of thinking strategies as options (e.g., P12 received analogical thinking, inspiration thinking, and critical review together). The plain notes (n=23, 12 sessions) were mostly used to pose a question or a single challenge, such as "*How about people with low vision?*". In general, the wizard's choices and rationales varied across designers despite a shared starting point and protocol, suggesting that proactive support needs to adapt to individual design trajectories rather than follow a fixed sequence.

5.1.3 How designers evaluated the usefulness of the perspectives. Designers rated more interventions as useful than not: 44 received a rating of (4 or 5) and 21 a rating of (1 or 2) ($M = 3.30$, $SD = 1.33$; see Figure 5 for the distribution). Table 2 lists the reasons why an intervention was considered useful or not. The most common reason was that the perspective **named something they had not considered** (13/20 sessions), most often a user, constraint, or risk the designer had not accounted for, followed by having **validated or developed their own thinking** (11/20). On the other side, an intervention was perceived as not useful either because it was **redundant or irrelevant**, saying what they already knew or had written (6/20), or because it was **not yet actionable** at that point in the work (6/20).

Regarding different categories of perspectives, user-persona perspectives were rated as most useful ($M = 3.69$, $SD = 0.60$), domain-expertise perspectives as second most useful ($M = 3.25$, $SD = 1.45$), and design-thinking perspectives rated as the least useful ($M = 3.08$, $SD = 1.46$), despite being the most often sent perspective. The frequent sending yet lower reception of design-thinking may reflect the challenge and effort of enacting a thinking strategy mid-task. P7 commented after receiving the reframing thinking strategy:

> "*I feel this comment is a little bit cliché. I think all designers know this; I just haven't reached this step yet.*" [P7]

In the post-session interview, we also asked participants to what extent they agreed that "*the AI perspectives made me realize points I might not have reached on my own, even given more time.*"[3] Their explanations reveal two different sources of value for proactive perspectives. Some introduced an angle the designer had not considered, while others

[3]We refer to this item later as *novelty*.

Primary code	Secondary codes	Example quote
Useful: what the designer perceived as valuable		
Named something they had not considered *(13/20 sessions)*	A genuinely unheld view: a user, a frame, or a fact they did not have | Something they had meant to do and dropped	"I hadn't really considered the perspective of a pedestrian … It will change my revision of the proposal." (P17)
Validated or developed their own thinking *(11/20 sessions)*	Validated a position they held | Developed a point they held with evidence or a next step	"it just really helped affirm an idea that I had" (P20)
Introduced a tension *(6/20 sessions)*	A risk or failure raised against the current idea	"it lets me know that red signals could cause panic rather than communicating safety" (P15)
Gave options by presenting alternatives together	Several perspectives side by side, holes and next steps at once	"some agents were able to point out holes in my proposal, and … other agents were able to provide next steps" (P4)
Not useful: what the designer perceived as not valuable		
Redundant or irrelevant *(6/20 sessions)*	Said what they already knew or had written | Repeated an earlier card, or read as a summary	"it's not useful because I have already taken down the notes while I was doing exploration" (P18)
Not yet actionable *(6/20 sessions)*	Solutions or cautions they could not use yet | Too thin to act on	"it's kind of pushing me to start looking at the solution when I wasn't [ready] …[‘] (P19)
Out of their standing to judge *(2/20 sessions)*	Couldn't understand it, or did not see it as theirs to judge	"But, like, I don't know… it's not my call to evaluate [a stakeholder]?" (P9)
Disagreed on substance *(2/20 sessions)*	Wrong on risk, cost, or feasibility	"It's certainly an interesting idea, but I feel like it presents a huge risk." (P17)

Table 2. Qualitative coding of usefulness explanations, with one illustrative example for each primary code. Ratings 4–5 were categorized as useful and ratings 1–2 as not useful; ratings of 3 were classified from their explanations.

articulated something the designer already believed. P9, for example, valued one intervention for confirming what they already thought. As P9 stated, "*You're agreeing with me.*" Yet confirmation of the designer's own thinking also appeared among explanations for low usefulness (Table 2). The difference seems to be whether the recognition did additional work: consolidating, or making articulable, a half-formed judgment, whereas merely restating a settled idea added little.

P9 described what separates a novel perspective from a merely useful one:

> *"When I was explicitly writing things down, I was really using more of my association memory … the AI coming up with new associations based on what I've already done would probably not be that surprising to me. But if it was going through and simulating out the ramifications of what I had said, that's episodic memory. I tend to be more surprised by the actual step-by-step carrying out stuff."* [P9]

5.2 RQ2: When is a good time for new perspectives to interrupt

The timing ratings were split. Forty-three interventions were rated (4 or 5) and 30 were rated (1 or 2), with only 8 at the midpoint ($M = 3.21$, $SD = 1.49$). Table 3 presents the coded reasons. Low timing ratings were most often associated with **occupied attention** (16/20 sessions), such as being in the middle of a thought or reading, followed by the designer's **own thinking not yet being ready for input** (6/20). High ratings were most often associated with a desire for external input (14/20), followed by having free attention (5/20).

5.2.1 Designer's openness to feedback was different across a session. Figure 6 shows how designers rated good and bad moments for each proactive perspective across their session. Grouping sessions by the shape of their timing-rating sequence, we descriptively grouped the sequences by their dominant shape into five recurring trajectories:

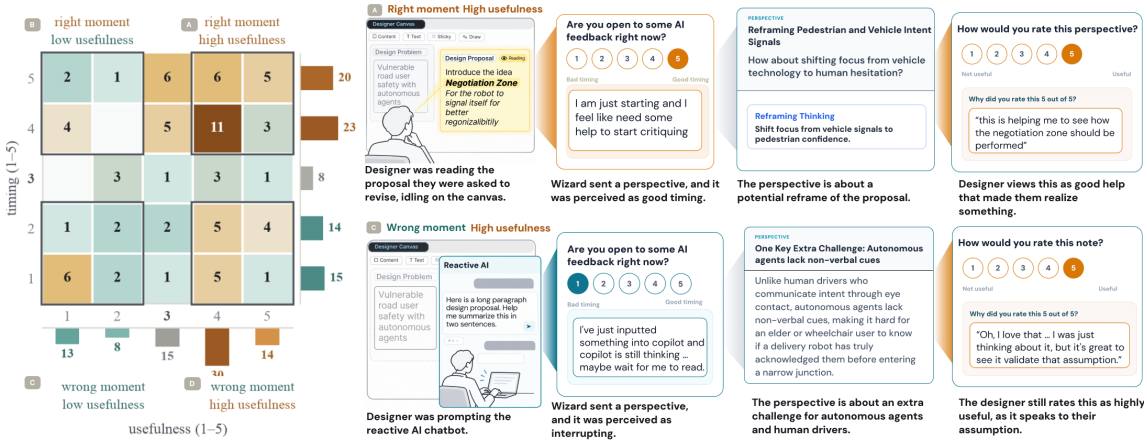


Fig. 5. Timing and usefulness ratings crossed for every rated intervention (left). The outlined corners mark the four combinations of a good or bad moment with high or low usefulness. The bars give the marginal counts, and ratings of 3 stay in the midpoint row or column. Right, two interventions traced from the designer's state to their ratings: one rated a good moment and highly useful (A), and one rated a bad moment yet highly useful (C).

- **Rises:** ends at least two points above its start, with at most one turn;
- **Peaks:** turns exactly once, at most once inside the session;
- **Falls:** ends at least two points below its start, with at most one turn;
- **Stays high:** averages at least 3.5 and never drops below 3;
- **Stays low:** averages at most 2.

Participants' retrospective interviews speak to them. P7, in the *Rises* trajectory, first experienced the interventions as intrusive:

> *"I thought it was an intrusion, like on Google Docs when they keep forcing Gemini on you when I didn't even have time to think about it"* [P7]

But once their own ideas had taken shape, they felt ready for "some external voice." P8, in the *Falls* trajectory, described the reverse, rating the first three interventions 5 and the last 2 once the proposal was finished: "*Normally, if I'm done with the proposal ... This is one of the moments I don't want any AI feedback … [I] feel like I am not doing a good job*". However, because ratings were collected only at wizard-selected intervention moments, these sequences are best treated as descriptive patterns rather than continuous trajectories of openness.

5.2.2 The wizard combined observations to decide when to weigh in, but timing remained uncertain. The wizard combined multiple observations, such as the canvas and the designer's exchanges with the reactive AI, to infer the designer's openness to feedback. Coding what the wizard's rationales said about the moment of intervention, the most common reasons were **responding to what the designer was working on**, **seeing the designer slow down**, and **one piece of work having just ended**. Each of these required the wizard to combine several observations to read the designer's openness.

Even these rich signals and contextual reading did not ensure successful timing: the same observable state could draw opposite timing ratings. Apparent idleness was particularly ambiguous. The platform flags the designer as idle when no keyboard or mouse input has reached the system for five seconds. Of the 24 interventions the log flagged as

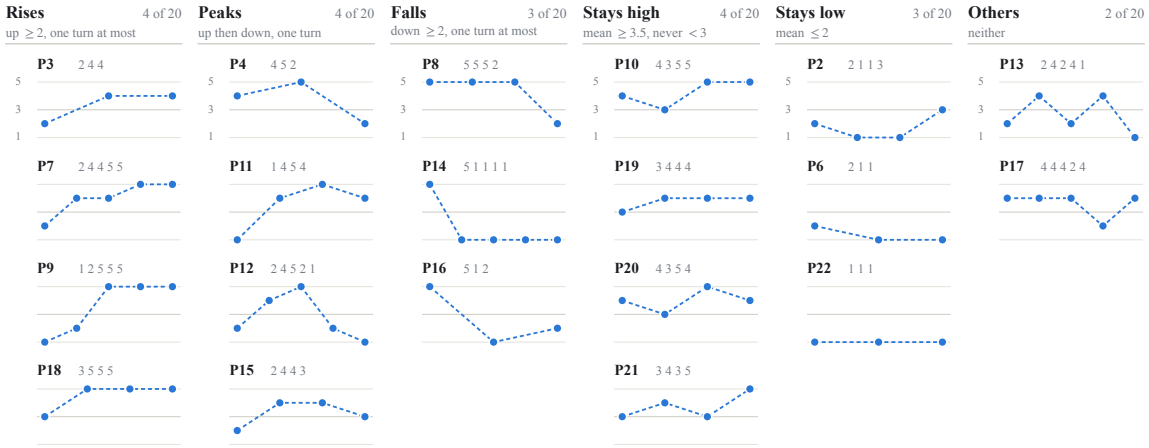


Fig. 6. Timing-rating sequences of the 20 sessions, grouped by shape. Each panel plots one session's timing ratings (1–5) in send order; the rule for each trajectory is given under its heading. Two sessions (right) fit none of the five trajectories.

idle, 17 were rated (4 or 5) and 6 were rated (1 or 2) regarding the timing. P16 rated one a 5 because "*I am just starting and I feel like I need some help to start critiquing*," whereas P22, also flagged as idle, rated a 1 because "*I'm still in the zone, I'm gonna keep going longer by myself in deep work.*"

Nor did visible activity rule out that they were stuck. P17 noted in the closing interview that "*if I keep typing stuff and then deleting what I write, [that is] probably a sign that I need a little bit of help verbalizing what I'm thinking*." The potentially informative signal was therefore not typing or idling alone, but the unfolding pattern of action and revision.

The same task could also draw divided timing ratings. Being engaged with the Copilot is one case: the 17 interventions that arrived while the trace showed the designer in the prompting panel split into 9 rated 4–5 and 8 rated 1–2. P19 rated one 4 because they "*[were] waiting for the Copilot to think*," while P11 rated one 1 because "*I've just inputted something into Copilot and Copilot is still thinking . . .*." Waiting for an AI response may indicate orientation toward assistance when the designer is stuck and wants feedback, but may simply be an interval in which the designer is trying to bounce ideas. An orchestrator must therefore interpret the current task in relation to both what the designer is trying to accomplish and where they are in the broader session.

Timing could also fail because the designer's activity changed while an intervention was arriving. They might begin writing, receive another AI response, or create a new note in the interim. We describe these cases as *cognitive collisions*, in which a moment judged to be open became occupied by concurrent activity before or as the perspective arrived. We identified 16 collisions across 13 sessions, and 11 of them were rated 1 or 2. In their reflections, the wizard consistently described the wait for image generation, which can take some time, as a touchpoint for intervening; the designer, however, could read the same wait either way. There were also observational misses, such as the wizard not noticing that the designer had begun typing. Richer context therefore improved the basis for a timing judgment but could not remove the uncertainty introduced by rapidly changing activity or by human oversight. To organize these patterns, we derived a framework from the data, presented next.

Primary code	Secondary codes	Example quote
Good timing: the designer was open		
Could use external input *(14/20 sessions)*	Stuck on what to do next | Exploring and wanting more ideas | Has an idea and wants it tested | Was about to ask the AI anyway	"I'm drawing a bit of a blank on what to do next." (P17)
Attention was free *(5/20 sessions)*	Just finished a piece of work | Waiting on a generation	"I am done with brainstorming and I'm ready to move on to the next task" (P8)
Interruption bandwidth *(3/20 sessions)*	Sparse enough to feel like a check-in	"it's just nice to feel like every now and then it's checking in. I feel like it's been infrequent" (P20)
Bad timing: the designer was not open		
Attention was occupied *(16/20 sessions)*	In the middle of a thought or a read | Already in an exchange with the Copilot	"I'm in the middle of forming a thought right now. My hesitation comes from me thinking rather than from me drawing a blank." (P17)
Own thinking not ready for input *(6/20 sessions)*	Wanted to keep working alone | Still framing the problem	"I'm still in the zone, I'm gonna keep going longer by myself in deep work." (P22)
Wrong point in the session *(4/20 sessions)*	Wanted it at a later stage | The work was already finished	"I think this should go near the end of my design session" (P6)
Interruption bandwidth *(2/20 sessions)*	No bandwidth left after the last one	"I wish it gave me more time between the last time I asked for feedback" (P15)

Table 3. Qualitative coding of explanations for timing ratings. Ratings 4–5 were categorized as good timing and ratings 1–2 as bad timing; ratings of 3 were classified from their explanations. Three ratings without a directional explanation are omitted. Here, n denotes coded interventions; session counts are non-additive across codes.

5.3 RQ3: What would an automated orchestrator need to take over from the wizard

5.3.1 Timing and content may require separate evaluation. Crossing the two ratings, fifteen interventions were rated 4 or 5 on usefulness despite a timing rating of 1 or 2, while seven were rated 4 or 5 on timing but 1 or 2 on usefulness (Figure 5). Descriptively, the paired ratings were weakly associated ($r = .20$, 95% CI $[-.02, .40]$). Because this estimate pools repeated observations within participants, we treat it only as descriptive evidence. Together with the mismatched cases above, it suggests that timing and usefulness need not move together in our task.

Even at a moment rated as bad timing, an AI perspective can still be perceived as useful, while a moment rated as good timing does not guarantee that a suggestion is good. The design implication is that a useful but mistimed intervention should prompt adjustment to the timing policy without penalizing its content, whereas a timely but unhelpful intervention should prompt changes to perspective selection or generation.

5.3.2 Designers envisioned future proactive AI to have more mixed initiative. In the post-session interviews, four designers said that the advantage of the proactive AI over a chatbot was that it already knew the board, and wanted it and the reactive Copilot to share that context. P11 explained: "I always wish that I don't have to re-explain myself… contextual knowledge is where AI tools can win, and that's where the beauty of proactive AI can serve." Eight designers also wanted a way to ask for a perspective at will, for example, a right-click on a sticky note asking the AI to "criticize me" (P10), or a request for a particular kind of perspective. As P12 put it, "having it pop up but also me being able to do it manually would be helpful." In other words, the proactive perspectives need not be system-initiated only: the same channel could also be triggered reactively, by the designer's own request.

Seven designers wanted the AI support fitted to the kind of help they need and to their own way of working. Designers wanted the proactive AI to learn their patterns of cognitive openness and also the perspectives needed. P8 would have the system "just watch people, how they do it" for the first few sessions, so that when "people reject the AI

response, AI learns, okay, this is not the right time to pop up… not just about the timing, but also the way of thinking," P6 asked it to distinguish big-picture thinkers, who need help with detail, from detail-focused ones, who need the reverse, and P20 would open with a quiz on "what type of brainstormer" the designer is. Personalization here is not a preference setting but the policy that decides when and how the orchestrator weighs in.

Six designers wanted to arrange in advance how perspectives should reach them, rather than judging each one as it arrived. P22 wanted the AI to be similar to how they would work with a human colleague—"we'd probably do a one-hour workshop, and then we go off and do deep work, and you're not allowed to interrupt me the entire day"—and would trade surprises for a schedule, saying "you've worked for an hour, now's the time to have a review session. Like, more predictability." Others would set the terms themselves, making proactive feedback a mode they enter. P6 put it by analogy to Figma's design and dev modes: "I need an ideation mode, I need someone to talk to, and then [other times] I'm like, don't bother me." P8 inverted the usual consent, proposing that a designer who knows they avoid critique should be able to say in advance, "even if I reject you every time, still, don't listen to me." None of these arrangements removes the system's initiative. They keep it noticing and return the reveal to the designer, which is the arrangement an automated orchestrator would have to honor: it may decide what is worth offering, but the terms on which an offer becomes an interruption are set by the person working.

5.3.3 Implications for future proactive orchestration.

Taken together, our findings point to several considerations for future proactive design support.

A three-layered framework to find good timing. Regarding choosing good timing, we derived three layered scales that can help infer a designer's openness to feedback: *session stage* describes the designer's position in the overall session; *task frame* describes the objective and object of the work currently underway; and *attention availability* describes whether the designer's attention is available in the next few seconds. We coded the designers' explanations of their timing ratings and the wizard's send rationales for all 81 interventions. For 79 of them, the designer's explanation (64) and/or the wizard's rationale (66) mentioned at least one of the three scales. Figure 7 shows the three scales. For each scale, it also lists the keywords associated with it and the number of sessions in which each appeared, followed by one intervention rated (1 or 2) and one rated (4 or 5) that share the same observable state.

- **Session stage** places an intervention along the broader arc of the session, from entering the problem to settling a proposal. Designers and the wizard referred to states such as *framing the problem*, *ideating*, *converging*, and *wrapping up* (Figure 7). These stages loosely resemble the diverging and converging structure of the double diamond model [15], but preferences for when to receive input varied across designers.
- **Task frame** captures the objective of the work currently underway. It appeared in 28 of the 81 designer timing reasons across 15 sessions and in wizard rationales across 11 sessions, including activities such as prompting the reactive AI or exploring an idea. As Section 5.2.2 shows, the same task could accompany both high and low timing ratings, suggesting that receptiveness depended on how an intervention related to the designer's current objective.
- **Attention availability** concerns whether the designer had attention available at the moment of delivery. Observable activities such as idling, typing, or reading were not sufficient on their own to determine openness. The wizard therefore often delayed delivery, holding 31 of 81 interventions in the queue across 15 sessions, with 14 held for at least a minute while waiting for a pause or transition.

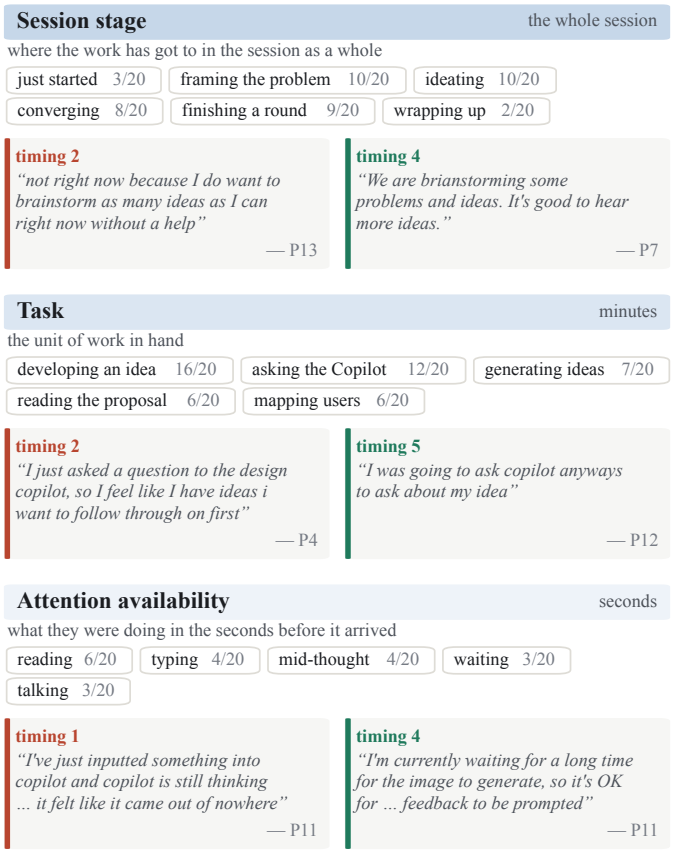


Fig. 7. Timing explanations at three nested scales. Each band gives the scale, what it means, and the states named at that scale, with the number of sessions (of 20) in which the designer or the wizard named each; below them, an intervention rated 1–2 and one rated 4–5 that share the same observable state, each labeled with the designer's timing rating.

Second, deciding What feedback to offer requires considering what a perspective adds to the designer's current thinking—for example, surfacing an overlooked concern, stress-testing an emerging idea, or developing a point that is not yet settled—rather than simply whether the feedback is relevant. An automated system could operationalize this by generating multiple grounded candidates and evaluating how each might contribute beyond what the designer has already considered.

Third, When and What may benefit from being treated as distinct judgments. Designers sometimes valued a perspective despite a low timing rating, while an apparently good moment did not ensure that the perspective itself would be useful. An automated orchestrator could therefore estimate receptivity and perspective value separately, and use the two judgments together when deciding whether to intervene, defer a perspective, or continue observing.

Appendix Table 5 extends these considerations into five study-informed dimensions of proactive orchestration and sketches candidate ways they might be operationalized. These are not intended as a validated automation policy, but as hypotheses and design directions for future systems to implement and test.

6 DISCUSSION

Our findings show why the timing of proactive feedback and the usefulness of its content need to be evaluated separately while informing a joint orchestration decision. Designers could find a perspective useful despite an unwelcome arrival, or welcome an intervention but find little value in its content (Section 5.3.1). These mismatches motivate design implications for when systems intervene, what they contribute, and how their behavior could adapt over time.

6.1 Interpreting good timing and coordinating design feedback

Hudson et al. [31] investigated predicting interruptibility from sensor-based observations, while Iqbal and Horvitz [32] examined how computer users suspend and resume ongoing tasks. Our findings complement these accounts by showing how designers explain apparently similar activity in different ways. Of the 24 interventions sent when the interface recorded the designer as idle, 17 were rated (4 or 5) on timing and 6 were rated (1 or 2) (Section 5.2.2). For P16, a pause accompanied a need for help beginning the critique. For P22, a pause accompanied a desire to continue working alone. Momentary activity therefore needs to be interpreted in relation to what the designer is trying to accomplish, rather than used as a sufficient trigger on its own.

The framework in Section 5.3.3 organizes this context through session stage, task frame, and attention availability. This connects to Chan et al. [8], who found that presenting semantically distant inspirations during productive ideation slowed idea generation relative to providing no inspirations. Our designers' accounts similarly distinguish wanting a new direction from wanting to continue developing one already underway. Read alongside the interruption literature, this suggests that timing feedback involves both finding an opportunity to redirect attention and judging whether such a redirection fits the current work.

Designers also wanted to establish when feedback could enter their work, rather than decide anew for every arrival. Participants proposed arrangements such as scheduled reviews, protected work periods, and modes for engaging with feedback (Section 5.3.2). [17] found that making feedback available throughout a visual design task supported greater reductions in design issues but could encourage reliance on feedback instead of holistic self-evaluation. Their participants chose when to request feedback; our study raises the complementary question of how designers can establish the terms of unsolicited intervention. We use *negotiable intervention* to describe arrangements in which the system can propose a contribution while the designer can set or revise when and how it may interrupt.

The distinction McFarlane [44] draws between immediate, negotiated, mediated, and scheduled interruption coordination provides a vocabulary for these alternatives. In a proactive design tool, they could allow the designer to accept an offer, defer it, dismiss it, or request a perspective directly, consistent with the guidance of Amershi et al. [3] on invocation, dismissal, and correction. The system could continue preparing perspectives during protected work periods without revealing them immediately. Because even an offer can interrupt, future work should compare immediate intervention, less intrusive offers, and scheduled review opportunities, examining both disruption and subsequent engagement with feedback.

6.2 Broadening and developing designers' thinking with in-situ AI perspectives

The wizard selected perspectives according to what they might contribute to the current work. In 17 sessions, the wizard used at least two perspective categories; user-persona perspectives were usually intended to introduce overlooked stakeholders, while domain-expertise perspectives often examined failures or missing disciplinary concerns (Section 5.1.2). This complements Perspectra, which lets users invite expert agents into an evolving discussion [40]. Our

study examines what happens when perspectives are proposed without an explicit request. Together, these observations suggest combining automated and proactive selection of potentially missing perspectives with opportunities for designers to request a particular angle when they can identify the help they need (Section 5.3.2).

Designers' explanations distinguish several forms of value, including broadening their thinking and developing an idea they already hold. In 13 sessions, usefulness explanations described a perspective that introduced something not previously considered or recovered an omission; in 11, they described validation or development of their own thinking (Table 2). Both can support design work. Confirmation helped when it clarified an emerging idea, but was redundant when it merely restated a settled point (Section 5.1.3). An orchestrator should therefore consider what a perspective adds to the designer's current thinking, rather than treating either novelty or agreement alone as evidence of usefulness. In our study, AI-enacted personas served as prompts for reflection rather than representations of stakeholders' lived experience. They should not substitute for stakeholder participation or empirical evidence about those experiences.

This distinction matters for evaluation. Remy et al. [50] recommend defining a creativity-support tool's intended contribution and evaluating it accordingly. If the aim is to broaden designers' thinking, a high usefulness rating does not establish that an intervention introduced a new consideration. Our retrospective novelty question addresses participants' belief that a perspective raised a point they might not have reached on their own (Section 4.4). Our study did not measure the originality of their resulting design. Future evaluations could examine perceived novelty alongside usefulness and the subsequent changes designers make, distinguishing whether feedback introduces a concern, develops an idea, or merely repeats it.

6.3 From situated traces to adaptive and personalized orchestration

Adapting proactive support requires learning both the help a designer values and the circumstances in which they want it. Seven participants explicitly wanted support tailored to their working patterns and the kinds of help they needed (Section 5.3.2). An adaptive system could begin with lightweight questions about these preferences and refine its behavior through subsequent requests, deferrals, and evaluations. The mixed-initiative personalization approach of Bunt et al. [7] combines adaptive recommendations with user control. For proactive feedback, this suggests letting designers inspect and revise the system's assumptions about their preferences, rather than interpreting every dismissal as evidence that proactive support is unwanted.

Personalized timing has been demonstrated in more structured tasks, such as how Liu et al. [39] used physiological and mouse-movement signals to adapt proactive LLM assistance during online surveys. For design work, the accounts in our study suggest that adaptation also needs to distinguish a desire for help from a desire to continue thinking alone. Our evaluation separates reported openness before content is revealed from perceived usefulness after engagement. Those judgments can inform different adjustments for when to offer a perspective and which content to prepare. A missing usefulness rating should not be treated as evidence that unseen feedback was unhelpful.

6.4 Limitations

The study examined twenty sessions using one system and one proposal-refinement task. The stakeholder-rich task may have made persona perspectives particularly relevant, while revising a supplied proposal may elicit different responses from developing a project in which the designer has greater ownership or expertise. The findings therefore identify patterns to investigate across other design activities, rather than establish their prevalence.

Two operators served as wizards in different sessions, so the second operator shows that another person could enact the protocol but does not establish agreement between operators or the performance of an autonomous system.

The wizards selected content, timing, and presentation together; the observations cannot isolate the effects of these choices. Records of delayed or withheld candidates capture why those choices were made, but cannot establish how designers would have responded had the candidates been sent at those moments. The study also does not establish an improvement in design outcomes relative to reactive assistance alone.

The evaluation procedure may have shaped the experiences it measured. Although collecting the timing judgment before revealing the feedback prevented its specific content from informing that rating, asking participants to rate and explain the arrival added a task that may have affected their attention and subsequent engagement [28]. Participants were expected to evaluate interventions, limiting conclusions about how often unsolicited feedback would naturally be noticed or ignored. Usefulness judgments were collected after engagement and cannot establish the value of feedback that designers did not inspect.

7 CONCLUSION

Using a Wizard-of-Oz study of 20 design sessions and 81 proactive interventions, we examined what makes AI-enacted perspectives useful and when designers welcome them. Timing and usefulness judgments could disagree, and designers' explanations placed openness in the context of session stage, task frame, and attention availability. Useful perspectives both introduced overlooked considerations and helped develop existing ideas. These findings motivate evaluating timing and usefulness separately while considering them together in orchestration, and investigating how perspective selection, timing, and presentation can adapt to designers over time.

Disclosure of Generative AI Use

We produced all original writing in this manuscript and used ChatGPT to assist with shortening and grammar checking. AI coding assistants (Claude Code and Codex) were used to write the scripts that generate the data figures and tables.

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A Participant Demographics

Participant identifiers were assigned sequentially during recruitment and range from P1 to P22. The session recording was missing for P1, and P5 did not complete the study session and was excluded from analysis. The analysed sample is therefore P2–P4 and P6–P22 ($n = 20$).

Table 4. Participant demographics.

ID	Age	Gender	Major / background	Education	Design study / practice
P2	25	Female	Design, HCI	Graduate degree	~4 years
P3	28	Female	Design engineering	Graduate degree	8 years
P4	19	Female	Digital media design	High-school degree	About 2 years
P6	35	Female	–	Graduate degree	11 years
P7	–	Female	HCI	–	5 years
P8	25	Female	HCI PhD	Graduate degree	3 years
P9	29	Male	Human–computer interaction	Graduate degree	5 years
P10	28	Female	Architecture and art history	Undergraduate degree	9 years
P11	22	Female	Master of human–computer interaction	Graduate degree	4 years
P12	25	Female	Human–computer interaction	Graduate degree	4 years
P13	23	Female	HCI	Undergraduate degree	4 years
P14	26	Male	–	Undergraduate degree	7 years
P15	23	Male	HCI	Graduate degree	3 years
P16	32	Female	Human–computer interaction	Undergraduate degree	3 years (study only)
P17	28	Male	HCI (Masters)	Undergraduate degree	7 years
P18	25	Prefer not to say	HCI (Masters)	Undergraduate degree	7 years
P19	35	Female	HCI (Masters)	Undergraduate degree	5+ years (systems); ~1 year (visual/UX)
P20	25	Male	HCI	Undergraduate degree	8 years
P21	23	Male	HCI (Masters)	Graduate degree	5–6 years
P22	26	Male	HCI	Graduate degree	3 years

B Design Task and Seed Proposal

The study began from the same seeded canvas for every participant. The canvas contained the design problem, an existing proposal to revise, and the required deliverable reproduced below. Participants could retain, modify, or entirely reframe the proposal; the task did not require them to accept its assumptions or solution direction.

B.1 Design problem shown to participants

Design Task: Improving Vulnerable Users' Safety in the Era of Autonomous Agents

Context & Problem

Vulnerable road users (e.g., elders, wheelchair users) must constantly interpret ambiguous situations in shared streets with increasing automated agents, such as autonomous vehicles and robots.

Right-of-way is often unclear, interactions with automated agents are unpredictable, and intent is not easily readable. This creates fear, hesitation, and high cognitive load, especially in tight or high-conflict spaces.

B.2 Existing proposal provided as a starting point

Design Proposal: The following is an existing proposal from a smart city institute.

Think of this as a starting point. You can build on it, revise it, or reframe it entirely.

Existing Proposal

Introduce lightweight **Negotiation Zones** in high-conflict areas (e.g., narrow sidewalks, blind corners, entrances), where interactions between pedestrians and automated agents are made more explicit and coordinated.

In these zones:

- Subtle **spatial cues** (e.g., ground markings, texture changes, or lighting) indicate that negotiation is needed
- Automated agents adopt **recognizable behaviors** (e.g., slowing down, slight lateral shift, pausing) to signal yielding or passing intent
- Optional **ambient signals** (e.g., soft audio or projected indicators) reinforce what the agent is about to do

Rather than enforcing strict rules, the system creates a shared moment where both parties can **anticipate, adjust, and coordinate** more comfortably.

This could be explored in a specific scenario such as:

- a narrow sidewalk where a wheelchair user and a delivery robot approach each other
- a blind corner where visibility is limited

The goal is to shift the interaction from **uncertain and reactive → legible and negotiable**, reducing hesitation and making the experience feel safer and more controlled.

B.3 Deliverable and interaction instructions

Deliverable: Please use this frame to present your proposed direction of change. Explain what should be kept, what should be changed, and why. You may also include supporting materials, such as references, images, sketches, diagrams, or other evidence. Everything else can go to the working space.

Note: You can respond by writing, copying and editing the proposal, or using the voice input button in the bottom-left corner.

C Perspective-Generation Prompt

The system dynamically inserted the selected perspective and current work context into the following prompt template. Angle brackets mark runtime values; line wrapping was adapted for print. Category-dependent explainability clauses are omitted for brevity.

Shared system prompt

You are generating design feedback for a wizard-selected perspective card. The only lens is the perspective card itself. Do not present the response as a mode or lead with agent-mode slogans. Use the perspective name, category, description, and microprompt as the main source of truth. Tie the response tightly to the brainstorm theme and current focus; never give generic design advice.

Write in plain, everyday language that a busy designer understands in one read. Use short sentences and concrete nouns and verbs. Avoid buzzwords, jargon, metaphors, and dramatic or poetic phrasing. State the point directly. Return only a JSON object, with no surrounding prose or Markdown fences.

Perspective name: *<name>*

Perspective category: *<category>*

Perspective description: *<description>*

Perspective microprompt: *<microprompt>*

Brainstorm theme: *<theme>*

Current focus/context: *<context>*

Canvas summary: *<design task and structured canvas context>*

Other agents in this batch are listed with their descriptions. Cover only the ground unique to this lens. Do not make points those lenses would naturally make. During unpacking, do not repeat or lightly rephrase points they have already made.

Stage 1—teaser

Produce a teaser the designer understands in one read:

- `title`: one declarative sentence of 6–10 words stating this lens's main takeaway for this design; not a question, topic label, metaphor, or wordplay.
- `teaser`: two short prose sentences, at most 40 words total, stating the point, its concrete consequence/so-what, and what it suggests doing.

Return only `{"title": "...","teaser": "..."}`.

Stage 2—unpack

Expand and substantiate the teaser while remaining consistent with it:

- `key_points`: exactly three actionable sentences, at most 18 words each;
- `rationale`: a 10–16-word fragment explaining why this lens reasons this way;
- `watch_outs`: exactly two risks or tensions, at most 16 words each; and
- `next_moves`: exactly two concrete next steps, at most 16 words each.

Never leave a field empty or add meta-commentary. Return only a JSON object with `key_points`, `rationale`, `watch_outs`, and `next_moves`.

Table 5. Five study-informed considerations for proactive perspective orchestration and candidate directions for operationalizing them. The right column illustrates possible automation mechanisms; these mechanisms were not evaluated in the present study.

What the study suggests	How might AI orchestrate proactively?
WHEN *Can the designer receive feedback now?* • Designers explained timing in relation to session stage, the task at hand, and attention availability. • The same observable state could correspond to either openness to feedback or a desire to continue working alone.	Combine signals across these levels with the designer's preferences when estimating receptivity. *context* ← Encode(*stage*, *task*, *moment*, *preference*) (*receptivity*, *confidence*) ← EstimateReceptivity(*context*)
WHAT *Is a perspective worth preparing?* • Designers valued perspectives that surfaced overlooked concerns, stress-tested current work, or developed emerging ideas. • Relevant feedback could still add little when it repeated what the designer had already considered.	Generate multiple grounded candidates and estimate what each might add beyond the designer's current thinking. *candidates* ← GeneratePerspectives(*context*) For each perspective p: *perspectiveScore*(p) ← Evaluate(context fit ↑, added value ↑, grounding ↑, redundancy ↓, effort ↓) *candidate* ← $\arg\max_{p \in candidates}$ *perspectiveScore*(p)
WHEN × WHAT *What follows from timing and content?* • A perspective could be useful despite an unwelcome arrival. • An apparently good moment did not guarantee useful content.	Estimate receptivity and perspective value separately, then use their combination to select an action. *Receptivity high*: Consider intervening / Continue observing *Receptivity low*: Consider deferring / Continue observing
HOW *How should the action be realized?* • Designers differed in how much initiative and interruption they wanted from proactive support. • Participants envisioned alternatives such as immediate feedback, deferred feedback, protected work periods, and on-demand requests.	Treat initiative and attention demand as design choices rather than assuming that every candidate should be immediately revealed. *Interaction*: observe | offer | reveal | none *Presentation*: cue | question | note | card | bundle *Disposition*: now | queue | withhold | cancel
HOW OFTEN *When should each step run again?* • Perspective preparation and receptivity judgments need not follow the same cadence. • Designers differed in how frequently they welcomed unsolicited feedback and when they wanted periods of uninterrupted work.	Allow candidate preparation to run asynchronously while adapting intervention cadence to the designer's behavior and preferences. **Operation** / **Candidate cadence** Prepare perspectives: Can run asynchronously as the work evolves. Evaluate candidates: Can be updated as the design context changes. Offer a perspective: Should adapt to observed receptivity and designer preferences.